INCH POUND
MIL-PRF-1/211F
26 June 1998
SUPERSEDING
MIL-E-1/211E
18 December 1974

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 3C24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triode.

See figure 1.

Mounting position: Vertical, base down or up.

Weight: 1.5-ounces nominal (42.5 grams).

ABSOLUTE RATINGS: C telegraphy.

Parameter:	F1	Ef	Eb	Ec	Ib	Ic	Pp
Unit:	MHz	V ac	kV dc	V dc	mA dc	mA dc	W
Maximum:	60	6.6	2.0	-500	75	25	25
Minimum:	- - -	6.0	- - -	- - -	- - -	- - -	- - -
Test conditions:	- - -	6.3	1.0	Adjust	35	- - -	- - -

GENERAL:

Qualification - required.

DISTRIBUTION STATEMENT A: Approved for public release, distribution is unlimited.

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TABLE I. Testing and inspection.

Inspection	Method	Conditions	Acceptance Level (Percent defective)	Inspection Level or Code	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1								
Peak emission	1231	eb = ec = 1,000 v	0.65	II	is	1.5	- - -	a
Electrode voltage (grid)	1261		0.65	II	Ec	-14	-24	V dc
Total grid current	1266	note 1	0.65	II	Ic	- - -	-5.0	µA dc
Filament current	1301		0.65	II	If	2.80	3.15	Aac
Conformance inspection, part 2								
Low-frequency vibration	1031	Rp = 2,000 Ω; Eb = 250 V dc; Ec/Ib = 10 mA dc	- - -	- - -	Ep	- - -	500	mV ac
Bump	1036	Angle = 25°	- - -	- - -	- - -	- - -	- - -	- - -
Power oscillation	1236	F = 100 MHz; Eb = 1,000 V dc; Ib = 80 mA dc; Ic = 25 mA dc	- - -	- - -	Po	45	- - -	W
Amplification factor	1316		- - -	- - -	Mu	22	28	- - -
Direct-interelectrode capacitance	1331	No shield	- - -	- - -	Cgp Cin Cout	1.4 1.4 0.1	1.8 2.2 0.3	pF pF pF
Conformance inspection, part 3								
Life test	- - -	Group C; Pp = 25 W; t = 500 hours	- - -	- - -	- - -	- - -	- - -	- - -
Life-test end point:	- - -							
Peak emission	1231		- - -	- - -	is	1.0	- - -	a

NOTE:

1. This test is to be the first test performed at the conclusion of the holding period.

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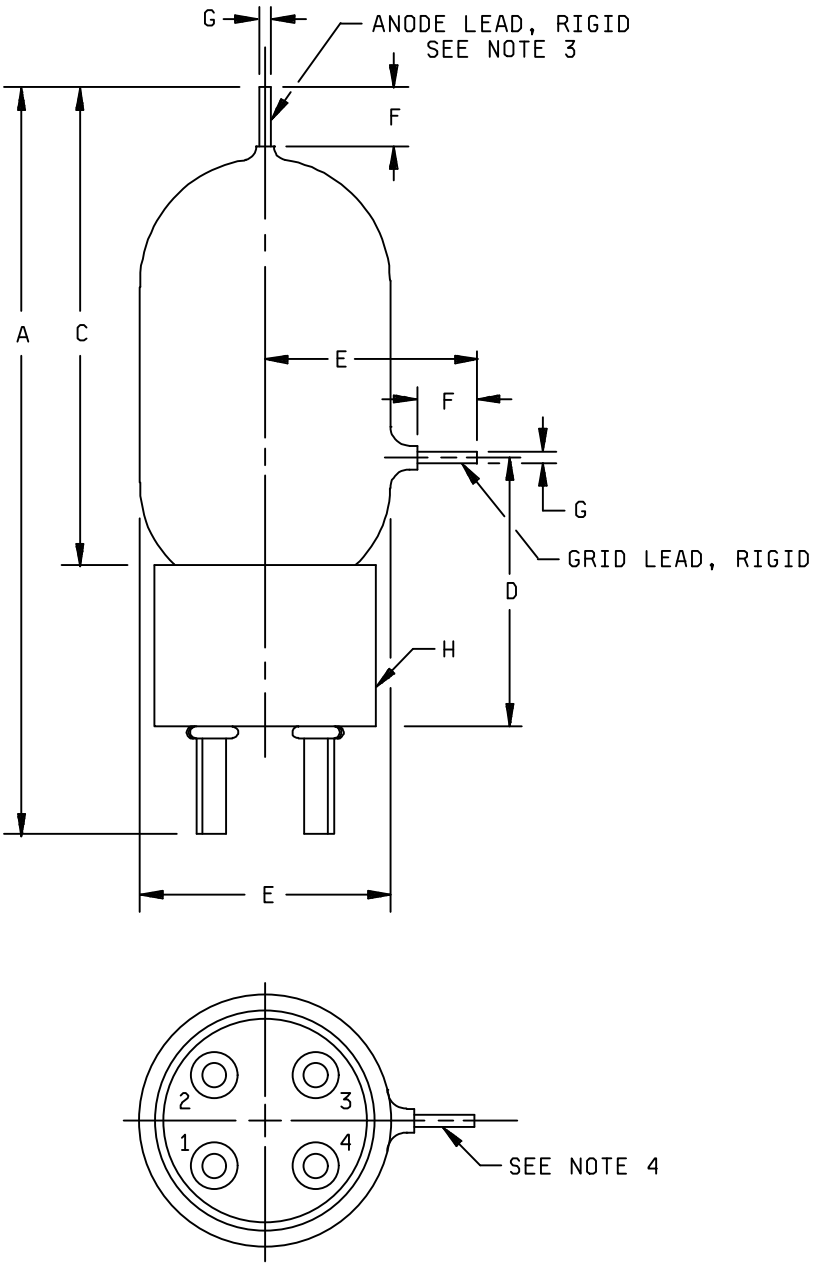

Pin connections	
Pin no.	Elements
1	f
2	nc
3	nc
4	f
Anode lead	a
Grid lead	g

FIGURE 1. Outline drawing of electron tube type 3C24.

See notes at end of Figure I.

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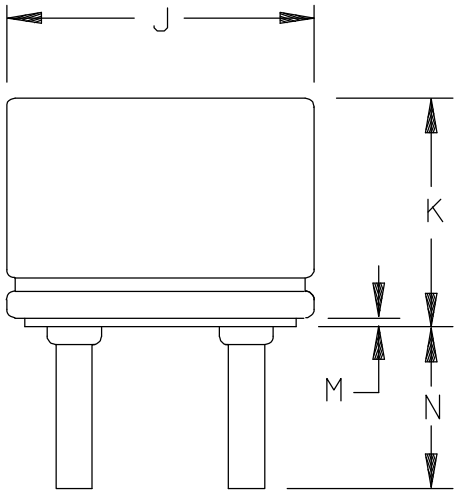

Alternate base, 4 pin, metal shell with ceramic wafer insert.

Dimensions		
Ltr	Minimum	Maximum
Conformance inspection, part 2		
A	4.000 (101.60)	4.375 (111.13)
B	1.313 (33.35)	1.438 (36.53)
C	3.404 (86.46)	
D	1.437 (36.50)	1.687 (42.85)
E	1.100 (27.94)	1.300 (33.02)
F	0.310 (7.87)	0.440 (11.18)
G	0.045 (1.14)	0.051 (1.30)
Conformance inspection, part 3 (see note 5)		
H	Base: A4-5 (see note 2)	
J	1.137 (28.88)	1.165 (29.59)
K	0.820 (20.83)	0.850 (21.59)
Reference dimensions		
M	0.032 (0.81)	
N	0.563 (14.30)	

NOTES:

1. For pin alignment, use gage GA4-1.
2. A metal shell base with ceramic wafer insert may be used.
3. Deviation of centerline of anode lead from centerline of base shall not exceed 0.125 (3.18 mm).
4. Deviation of centerline of grid lead from centerline of base shall not exceed 3°.
5. These dimensions shall be tested annually.

FIGURE 1. Outline drawing of electron tube type 3C24 - Continued.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 11, 17, 99

Preparing activities:
DLA - CC

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